

ApexPlanet Data Analytics Internship

Final Project Report

Student Name: Bandari Laxmi

College: [Sreenidhi Institute of Science and Technology]

Branch: B.Tech – Information Technology

Internship Organization: ApexPlanet Software Pvt. Ltd.

Internship Domain: Data Analytics

Project Title: Sales Data Analysis Using Python, SQL and Power BI

Dataset Used: Sample Superstore Dataset

Submission Date: July 2026

Executive Summary

Project Objective

The primary objective of this project was to analyze the Sample Superstore sales dataset and extract meaningful business insights using data analytics techniques. The project covered the complete analytics workflow, including data cleaning, exploratory data analysis (EDA), SQL-based data extraction, interactive dashboard creation in Power BI, statistical analysis, customer segmentation, and predictive modeling.

The goal was to convert raw business data into useful information that supports better decision-making and demonstrates practical data analytics skills.

Project Workflow

The project was completed in five stages:

Task 1 – Data Cleaning and Exploratory Data Analysis (EDA)

- Installed the required Python libraries.
- Imported the Superstore dataset into Jupyter Notebook.
- Removed duplicate records.
- Handled missing values.
- Corrected data types.
- Performed exploratory data analysis.
- Created charts using Matplotlib and Seaborn.
- Identified important trends and patterns.

Task 2 – SQL for Data Extraction

- Imported the cleaned dataset into SQLite.
- Executed SQL queries for business analysis.
- Used SELECT, WHERE, GROUP BY, ORDER BY, HAVING and JOIN operations.

- Connected Python with SQLite using SQLAlchemy.
- Answered business questions through SQL.

Task 3 – Data Visualization and Dashboarding

- Developed an interactive dashboard using Power BI.
- Added KPI cards for Sales, Profit, Orders and Customers.
- Designed line charts, bar charts, donut charts and filters.
- Built an executive dashboard for business reporting.

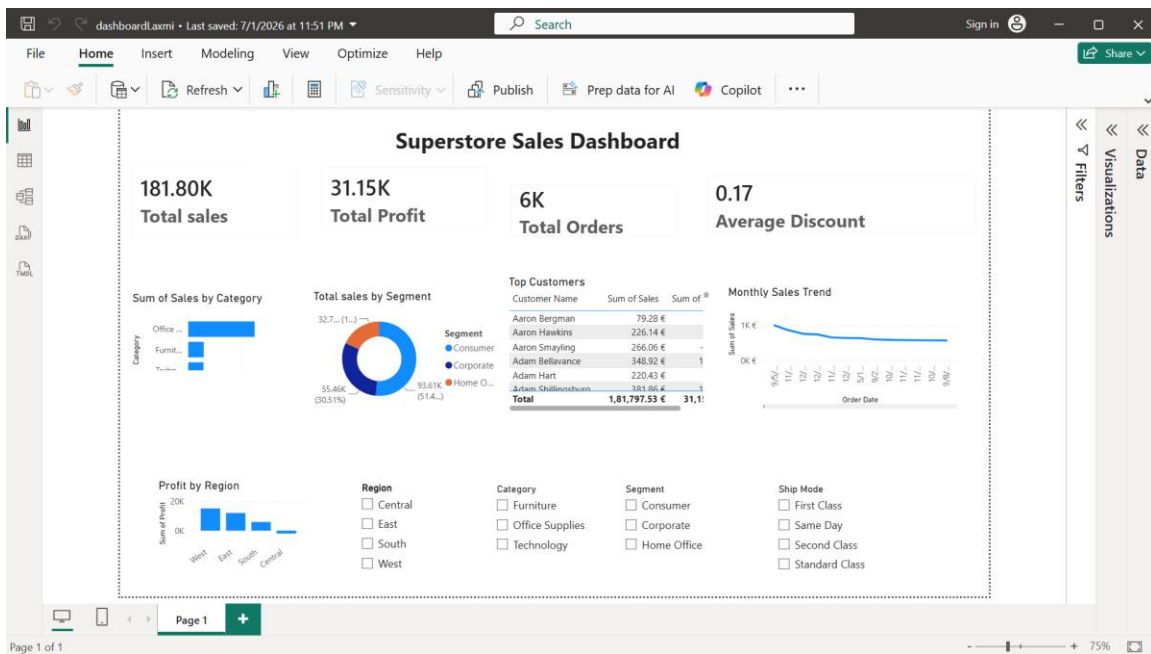
Task 4 – Advanced Analytics

- Performed descriptive statistical analysis.
- Conducted customer segmentation using K-Means clustering.
- Built a predictive model using Machine Learning.
- Evaluated model performance using appropriate metrics.

Task 5 – Final Report and Automation

- Prepared the final project report.
- Created an automation script using Python.
- Generated KPI reports automatically.
- Organized the project for final GitHub submission.

Dashboard Screenshot



Top Five Business Insights

- The Technology category generated the highest overall sales among all product categories.
- The West region contributed the highest revenue compared with other regions.
- Office Supplies generated comparatively lower profit margins despite high sales.
- Monthly sales increased significantly during festive and year-end periods.
- A relatively small number of customers contributed a large percentage of the total sales, indicating valuable high-spending customers.

Business Recommendations

- Increase inventory and promotional campaigns for high-performing Technology products.
- Improve marketing strategies and customer engagement in lower-performing regions to increase revenue.
- Launch customer loyalty and retention programs for high-value customers to encourage repeat purchases.

Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- SQLite
- SQLAlchemy
- Power BI
- Scikit-learn
- Jupyter Notebook
- Git & GitHub

Project Outcome

This internship project provided practical experience in the complete data analytics lifecycle. It enhanced skills in data cleaning, SQL querying, visualization, dashboard development, statistical analysis, machine learning, automation, and project documentation. The final dashboard and analytical findings demonstrate how business data can be transformed into actionable insights that support informed decision-making.

Conclusion

The successful completion of this project strengthened my understanding of real-world data analytics workflows. By integrating Python, SQL, and Power BI, I developed the ability to clean data, analyze business performance, create interactive dashboards, automate reporting tasks, and

communicate findings effectively. This internship has significantly improved my technical and analytical skills and prepared me for future opportunities in data analytics and business intelligence.